

COS 433 Problem Set 2

Due date: Wednesday, 10/7/2026 at 11:59pm. See [the course website](#) for our submission and grading policies.

Proof citations. Your proofs may assume without proof any fact stated in lecture or stated on the problem sets. All other statements – including results from textbooks or other sources that were not mentioned in lecture – should be proved in your problem set write-up.

Grading Policy. A total score of 100 points constitutes full credit for the problem set. Each problem is worth 25 points. You are not expected to complete all of the problems. Feel free to work on whichever problems you prefer!

1 COS 433 Problems

Problem 1: Multi-message security (25 points)

For this problem, you may assume any of the following: PRGs exist, PRFs exist, or CPA-secure secret-key encryption schemes exist.

- (a) Construct a secret-key encryption scheme (Enc, Dec) that is IND-secure but is not IND-CPA-secure. Prove that your scheme satisfies these properties.
- (b) Construct a secret-key encryption scheme (Enc, Dec) that is IND-secure for $t = 2$ messages but not IND-secure for $t = 3$ messages. Your scheme should support messages of length at least λ using a key of length λ .

Hint in the footnote below.¹

Problem 2: Double Encryption (25 points)

- (a) Assume that Enc is a secret-key encryption scheme with IND security. Is

$$\text{Enc}_1((k, k'), m) = \text{Enc}(k, \text{Enc}(k', m))$$

IND-secure? Here $k, k' \leftarrow \{0, 1\}^\lambda$ are two independently sampled keys. What if we reuse the same key, *i.e.*

$$\text{Enc}_2(k, m) = \text{Enc}(k, \text{Enc}(k, m)).$$

¹First try to solve the problem allowing for a larger secret key.

(b) Now assume that that Enc is CPA-secure. Are Enc_1 and Enc_2 CPA-secure?

Problem 3: The GGM PRF (25 points)

In class, we constructed a PRF family $\{\{F_s : \{0,1\}^n \rightarrow \{0,1\}^\lambda\}_{s \in \{0,1\}^\lambda}\}_{\lambda \in \mathbb{N}}$ using a binary tree of PRG evaluations. However, in this construction, we only defined how to evaluate the PRF on inputs of length *exactly* n .

In this problem, we consider how to build a PRF family with domain $\{0,1\}^*$. That is, we want to be able to evaluate $F_s(x)$ for *any string* x , no matter how long.

(a) Suppose we define $F_s(x) = s_x$ exactly as in class, for all $x \in \{0,1\}^*$. Explain why this candidate PRF is insecure.

(b) Show how to modify the GGM construction in a way that results in a secure PRF family with domain $\{0,1\}^*$. You do not need to prove your construction is secure, but your construction should be secure by a similar argument to what you have seen in class.

Problem 4: MAC Security (25 points)

Let $\text{MAC} = (\text{Sign}, \text{Verify})$ be a message authentication code. We say that MAC is *strongly secure* if for any polynomial-time adversary $\mathcal{A}$, there exists a negligible function $\epsilon(\lambda)$, such that

$$\Pr \left[G_{\mathcal{A}}(1^\lambda) = 1 \right] \leq \epsilon(\lambda)$$

where $G_{\mathcal{A}}(1^\lambda)$ is the following game:

- Sample $\text{sk} \leftarrow \{0,1\}^\lambda$.
- Run $(m, \sigma) \leftarrow \mathcal{A}^{\text{Sign}(\text{sk}, \cdot), \text{Verify}(\text{sk}, \cdot)}(1^\lambda)$, where the adversary $\mathcal{A}$ is given oracle access to both $\text{Sign}(\text{sk}, \cdot)$ and $\text{Verify}(\text{sk}, \cdot)$ (instead of given only $\text{Sign}(\text{sk}, \cdot)$). Let Q_1 denote the queries made by $\mathcal{A}$ to the oracle $\text{Sign}(\text{sk}, \cdot)$, and Q_2 denote the queries made by $\mathcal{A}$ to the oracle $\text{Verify}(\text{sk}, \cdot)$.
- The game $G_{\mathcal{A}}(\lambda)$ outputs 1 if and only if $m \notin Q_1$ and $\text{Verify}(\text{sk}, m, \sigma) = 1$.

Compared to regular security, the adversary in this game is more powerful: they can query *either* a signing oracle *or* a verification oracle.

Suppose that MAC is a deterministic scheme: $\text{Sign}(k, m)$ is a deterministic algorithm, and $\text{Verify}(k, m, \sigma)$ checks that $\sigma = \text{Sign}(k, m)$. Prove (via a formal security reduction) that MAC is secure if and only if MAC is strongly secure.

Hint in the footnote below.²

Bonus: What if the scheme is allowed to be randomized?

²During the execution of $\mathcal{A}$, what happens if $\mathcal{A}$ ever gets $\text{Verify}(\text{sk}, m', \sigma') = 1$ by querying the oracle $\text{Verify}(\text{sk}, \cdot)$ where $m' \notin Q_1$?

Problem 5: Feistel Networks and Inverse-Query PRPs (25 points)

Let $\text{PRP} = \{\{P_k : \{0, 1\}^{n(\lambda)} \rightarrow \{0, 1\}^{n(\lambda)}\}_{k \in \{0, 1\}^{l(\lambda)}}\}_{\lambda \in \mathbb{N}}$ denote a family of pseudorandom permutations. We say that PRP is *inverse-query secure* if for all polynomial-time oracle algorithms $A^{f_1(\cdot), f_2(\cdot)}$, the following distributions are computationally indistinguishable:

$$\left\{ A^{P_{\mathbf{k}(\cdot)}, P_{\mathbf{k}(\cdot)}^{-1}(\cdot)}(1^\lambda) \text{ for } \mathbf{k} \leftarrow \{0, 1\}^{l(\lambda)} \right\} \approx_c \left\{ A^{P(\cdot), P^{-1}(\cdot)}(1^\lambda) \text{ for } P \leftarrow \text{Permutations}(\{0, 1\}^{n(\lambda)}) \right\}.$$

That is, in this security definition, we give the adversary oracle access to both the permutation P and its inverse P^{-1} .

Let $\text{PRP}_{r=3, F} = \{\{P_{k_1, k_2, k_3} : \{0, 1\}^{2 \cdot n(\lambda)} \rightarrow \{0, 1\}^{2 \cdot n(\lambda)}\}_{(k_1, k_2, k_3) \in \{0, 1\}^{3 \cdot \lambda}}\}_{\lambda \in \mathbb{N}}$ be the pseudorandom permutation family built from a Feistel network with $r = 3$ rounds and pseudorandom function $F : \{0, 1\}^\lambda \times \{0, 1\}^{n(\lambda)} \rightarrow \{0, 1\}^{n(\lambda)}$, as seen in class. Namely, for $(x, y) \in \{0, 1\}^{2 \cdot n(\lambda)}$ we have:

$$\begin{aligned} P_{k_1, k_2, k_3}(x, y) &= (x_4, y_4), \text{ where} \\ (x_1, y_1) &= (x, y) \\ (x_2, y_2) &= (x_1 \oplus F_{k_1}(y_1), y_1) \\ (x_3, y_3) &= (x_2, y_2 \oplus F_{k_2}(x_2)) \\ (x_4, y_4) &= (x_3 \oplus F_{k_3}(y_3), y_3) \end{aligned}$$

Compute the following $(x_4, y_4) := P_{k_1, k_2, k_3}(x, y)$, $(x'_4, y'_4) := P_{k_1, k_2, k_3}(x \oplus r, y)$ and $P_{k_1, k_2, k_3}^{-1}(x'_4 \oplus r, y'_4)$ in terms of $x, y, r \in \{0, 1\}^{n(\lambda)}$. Show that $\text{PRP}_{r=3, F}$ is not inverse-query secure.

Problem 6: PRF Transformations (25 points)

Let F be a pseudorandom function such that, for $s \in \{0, 1\}^\lambda$, each F_s maps λ -bit inputs to λ -bit outputs. We would like to extend the domain of our PRF to inputs of length $n(\lambda) > \lambda$. We attempt to do so, by constructing F' as follows:

- Let A be a $\lambda \times n$ binary matrix chosen uniformly at random.
- For each $s \in \{0, 1\}^\lambda$, $F'_{A, s}(x) = F_s(Ax \pmod{2})$

(a) Suppose that the adversary knows A . Is this scheme secure? Prove your answer.

(b) Suppose that the adversary does not know A ; that is, we include A as part of the PRF seed. Is this scheme secure? Prove your answer.

Problem 7: PRF security (25 points)

In class, we analyzed the security of the Goldreich-Goldwasser-Micali (GGM) construction of a PRF family from a length-doubling PRG. In this problem, you will give a *different* proof of security of the same construction.

For this proof, we will define a notion of PRFs *with partial evaluation keys*. Specifically, we say that a (syntactic) PRF family $\{F_s : \{0, 1\}^{n(\lambda)} \rightarrow \{0, 1\}^{m(\lambda)}\}_{s \in \{0, 1\}^\lambda, \lambda \in \mathbb{N}}$ can be “punctured” if there exists an efficient algorithm $\text{KeyGen}(s, x) \rightarrow k_x$ and an efficient algorithm $\text{Eval}(k, x')$ with the following properties:

- **Correctness:** For all $x \neq x' \in \{0, 1\}^n$, $\text{Eval}(k_x, x') = F_s(x')$ with probability 1. We will use the convention that $\text{Eval}(s, x) = F_s(x)$ also describes “ordinary” PRF evaluation.
- **Pseudorandomness:** For all $x \in \{0, 1\}^n$, the distribution

$$\left\{ s \leftarrow \{0, 1\}^\lambda, k_x \leftarrow \text{KeyGen}(s, x) : (k_x, F_s(x)) \right\}$$

is computationally indistinguishable from the distribution

$$\left\{ s \leftarrow \{0, 1\}^\lambda, k_x \leftarrow \text{KeyGen}(s, x), u \leftarrow \{0, 1\}^m : (k_x, u) \right\}.$$

In other words, the function value $F_s(x)$ is pseudorandom *even given* k_x , which can evaluate F_s on *all other inputs*.

- (a) Prove that the GGM construction can be punctured, assuming that the underlying PRG is secure.
- (b) Prove that *any* family of functions $\{F_s : \{0, 1\}^n \rightarrow \{0, 1\}^m\}_{s \in \{0, 1\}^\lambda, \lambda \in \mathbb{N}}$ that can be punctured (as above) is a secure PRF family according to the definition from class.